National University of Computer and Emerging Sciences

Lab Manual 06
Programming Fundamentals

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Activities:

Note: Include input validation in programs wherever applicable. Don't use any build-in function.

1. *Sum of Odd and Even Digits*

   Write a program that reads a number from the user and calculates the sum of its odd and even digits separately using for loop.
   Example Input: N = 453672
   Output: Sum of odd digits = 15, Sum of even digits = 12

2. *Geometric Series Sum*

   Write a program that calculates the sum of a geometric series. Ensure using for loop.
   $$S = 1 + r + r^2 + r^3 + \cdots + r^{n-1}$$
   where the user provides the values of r (common ratio) and n (number of terms).
   Example Input: r = 2, n = 4
   Output: S = 15

3. *Generating Armstrong Numbers*

   Write a program that finds all Armstrong numbers between 100 and a user-specified upper limit N. A number is an Armstrong number if the sum of its digits raised to the power of the number of digits equals the number itself. For example: 153 is Armstong number as $1^3 + 5^3 + 3^3 = 153$ & 1634 is also Armstong number as $1^4 + 6^4 + 3^4 + 4^4 = 1634$
   Example Input: N = 500
   Output: Armstrong numbers: 153, 370, 371

4. *Floyd's Triangle*

   Write a program that generates Floyd's triangle for a given number N. Floyd's triangle is a triangular array of natural numbers where each row contains a continuous sequence of integers.
   Example Input: N = 5
   Output:

   ```
   1
   2 3
   4 5 6
   7 8 9 10
   11 12 13 14 15
   ```

5. *Right-Angled Triangle Pattern*

   Write a program that prints a right-angled triangle pattern of asterisks (*), like the one following. The user should specify the number of rows for the triangle.
   Example Input: Enter the number of rows: 5
   Output:

   ```
       *
      * *
     * * *
    * * * *
   * * * * *
   ```

6. *Unique Pairs of Numbers*

   Write a program that generates all unique pairs of numbers between 1 and N using nested for loops. For each pair (i, j), the output should only include pairs where i < j.
   Example Input: N = 5
   Output: (1, 2), (1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5), (3, 4), (3, 5), (4, 5)

7. *Collatz Conjecture*

   Write a program that generates the Collatz sequence for a user-specified positive integer N. The sequence is generated as follows:

   - If N is even, divide it by 2
   - If N is odd, multiply it by 3 and add 1

   The process is repeated until N becomes 1.

   Example Input: N = 6

   Output: 6 -> 3 -> 10 -> 5 -> 16 -> 8 -> 4 -> 2 -> 1

8. *Prime Factorization*

   Write a program that finds all prime factors of a user-specified number N.

   Example Input: N = 56

   Output: Prime factors = 2, 2, 2, 7

9. *Bank Account Management*

   Write a program that simulates a simple bank account system. The user can deposit or withdraw money, and the program should display the current balance after each operation. The user can-1 to quit the program. No withdrawal with appropriate message, if the withdrawal amount is greater than the current balance.

   Console window as an Example: (Considering initial balance as 0)

   Bank Account Menu:
    1. Deposit
    2. Withdraw
    -1. Quit

   Choose an option: 1
   Enter amount to deposit: 500
   Deposited: 500
   Current balance: 500

   Choose an option: 2
   Enter amount to withdraw: 200
   Withdrawn: 200
   Current balance: 300

   Choose an option: -1
   Final balance = 300

10. *Average Grade Calculation*

    Write a program that calculates the average grade for N students. Each student's grade must be between 0 and 100. Use a for loop to collect the grades. Moreover, do validate the input. Additionally, the program should track the number of valid grades entered and display this count along with the average.